Problem Set 2: Volatility

MGT 927

We use daily CRSP data from WRDS for 2 January 1990 to 31 December 2020, which gives 7,812 trading days. The market series is the CRSP value-weighted return (`vwretd`) and MSFT is its CRSP holding-period return, both in percent per day. We code every likelihood by hand and maximize it numerically. Standard errors are the Bollerslev–Wooldridge robust sandwich, and the point estimates agree with the `arch` Python package to three decimals. The code is in `volatility_ps.py`.

	Mean	S.d.	Ann. vol	Skew	Excess kurt.	Min	Max
Market	0.044	1.13	17.9%	−0.32	11.0	−11.8	11.5
MSFT	0.102	2.02	32.1%	0.23	6.5	−15.6	19.6

Question 1

(a) Autocorrelations of returns and squared returns

Before looking at the data we expected the market to show the stronger volatility clustering. Write MSFT's return as $r_{i,t} = \beta r_{m,t} + e_{i,t}$. The part of volatility that persists for weeks or months comes from economy-wide conditions such as recessions, financial stress and uncertainty about policy, and the index is exposed to all of it. MSFT carries the same market component plus firm-specific news: earnings announcements, product news and, in this sample, an antitrust case. That kind of news tends to arrive as a single large move that is not followed by more large moves. Each such day adds a big value of $r_{i,t}^2$ that has no counterpart in the days around it, so it raises the variance of squared returns without raising their autocovariance, and the measured autocorrelation falls. Diversification strips this component out of the index. For returns themselves we expected autocorrelations close to zero in both series, since predictable daily returns in liquid stocks would be traded away.

The data confirm the prior. Squared demeaned returns are positively autocorrelated at all 20 lags in both series, so both show volatility clustering, but the market's clustering is much stronger. Its autocorrelations average 0.256 over the 20 lags against 0.103 for MSFT, and its Ljung–Box $Q(20)$ is 11,056 against 1,853. The market's autocorrelations also fade more slowly, keeping 57% of their lag-1 value at lag 20 (0.176 against 0.311), while MSFT's keep 26% (0.048 against 0.182). The same ordering holds for absolute returns, which are much less sensitive to a few extreme days, so the result does not come from outliers.

Two further facts support the mechanism behind the prior. First, the firm-specific part of MSFT's return, the residual from a regression on the market, has squared-return autocorrelations that average only 0.062. MSFT's autocorrelations sit between that value and the market's, which is what one would expect if they are a blend of a strongly clustered market component and a weakly

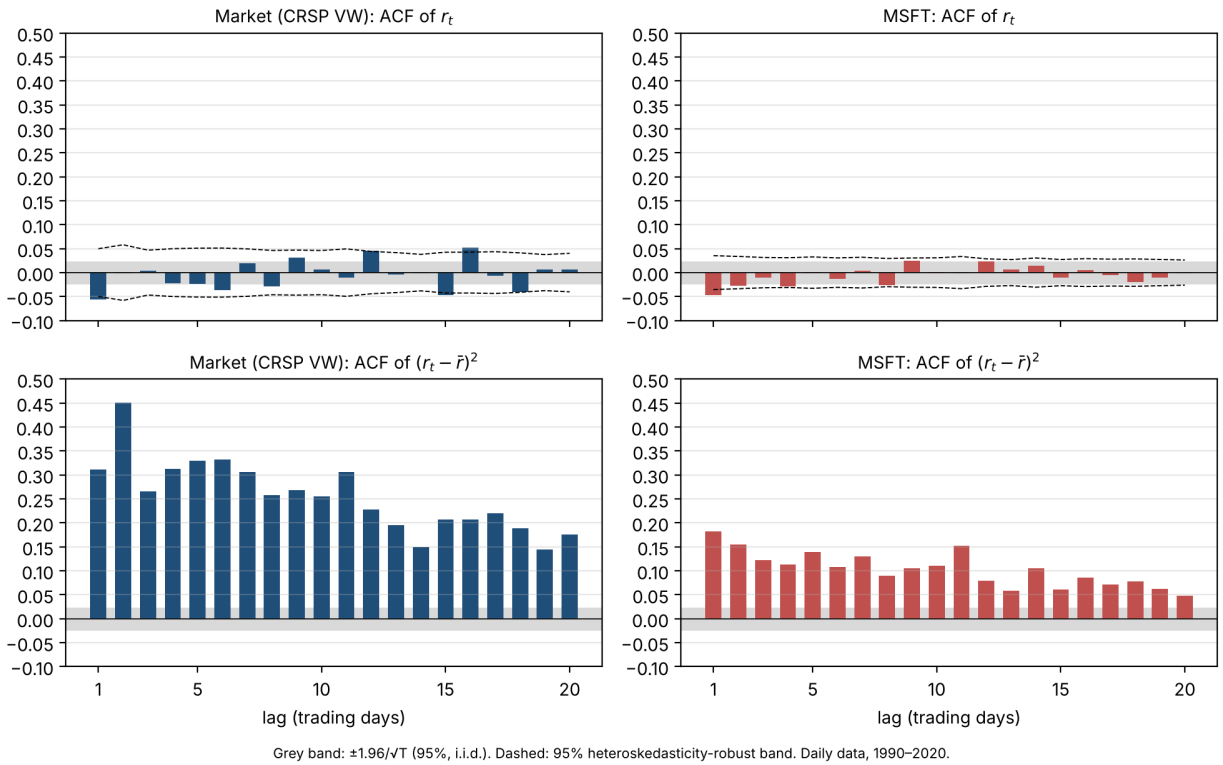


Figure 1: Autocorrelations at lags 1–20. Grey band: i.i.d. 95% band $\pm 1.96/\sqrt{T}$. Dashed: heteroskedasticity-robust 95% band.

clustered firm component. Second, MSFT's largest moves were firm-specific jumps: +19.6% on 19 October 2000, −14.5% on 3 April 2000 and −15.6% on 24 April 2000, days on which the market moved by less than 4%. The unconditional excess kurtosis is actually higher for the market (11.0 against 6.5), but that is because volatility clustering itself fattens the unconditional tails. After the GARCH model in part (b) removes the clustering, MSFT's standardized residuals have kurtosis 8.8 against 4.9 for the market, which is the signature of jumps that the volatility state does not predict.

The return autocorrelations are small, as expected. None exceeds 0.06 in absolute value. Against the usual $\pm 1.96/\sqrt{T}$ band, 10 of the market's 20 lags and 6 of MSFT's look significant, but that band assumes constant variance and is too narrow here. With a band that allows for time-varying variance, 4 market lags and 1 MSFT lag remain significant, and the largest coefficient, −0.056 for the market at lag 1, implies that less than 0.4% of the return variance is predictable from the previous day.

(b) Gaussian GARCH(1,1) by maximum likelihood

The model is

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \quad z_t \overset{iid}{\sim} N(0,1), \qquad \sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2,$$

with $\omega > 0$, $\alpha, \beta \geq 0$ and $\alpha + \beta < 1$. Conditional on $\mathcal{F}_{t-1}$, $r_t \sim N(\mu, \sigma_t^2)$ and σ_t^2 is a known function of $\theta = (\mu, \omega, \alpha, \beta)$ and past returns. The log likelihood is therefore the sum of the one-step

conditional normal log densities,

$$\mathcal{L}(\theta) = -\frac{1}{2}\sum_{t=1}^{T}\left[\log(2\pi) + \log\sigma_t^2(\theta) + \frac{(r_t-\mu)^2}{\sigma_t^2(\theta)}\right], \qquad \hat{\theta} = \arg\max_{\theta}\mathcal{L}(\theta),$$

where the variance recursion starts from $\sigma_1^2 = T^{-1}\sum_t (r_t-\mu)^2$.

	μ	ω	α	β	$\alpha+\beta$	Half-life	Long-run vol	$\log L$
Market	0.071	0.0175	0.112	0.873	0.985	47	17.4%	−10,110.2
	(0.009)	(0.0035)	(0.012)	(0.012)	(0.005)			
MSFT	0.116	0.066	0.079	0.906	0.985	47	33.9%	−15,649.6
	(0.018)	(0.036)	(0.025)	(0.030)	(0.009)			

Table 1: GARCH(1,1) estimates; robust s.e. in parentheses. The s.e. of $\alpha+\beta$ uses the delta method with the full covariance matrix. Half-life (days) $= \log 0.5/\log(\alpha+\beta)$ for deviations of the variance from its long-run level. Long-run vol $= \sqrt{252\,\omega/(1-\alpha-\beta)}$.

Both fits have the usual daily-data shape. A return surprise moves next-day variance only a little (α around 0.1), but the variance remembers its past almost completely (β around 0.9), so $\alpha+\beta = 0.985$ and a deviation of the variance from its long-run level takes about 47 trading days to shrink by half. Whether persistence is significantly below one depends on the standard error of the sum, and that standard error cannot be read off the two individual ones. The estimates of α and β are strongly negatively correlated (-0.93 for the market and -0.97 for MSFT), because the likelihood can trade a higher α against a lower β with little change in fit. With the covariance included, the standard error of $\alpha+\beta$ is 0.0046 for the market and 0.0086 for MSFT. A one-sided test of the integrated null $\alpha+\beta = 1$ rejects clearly for the market ($t = 3.2$) and only marginally for MSFT ($t = 1.7$, $p \approx 0.045$), so MSFT's long-run variance is estimated much less precisely.

The two series differ in how they reach the same persistence. The market's variance reacts more to each surprise ($\alpha = 0.112$ against 0.079), which is the model's version of its higher squared-return autocorrelations in part (a), while MSFT relies more on its own lagged variance. The implied long-run volatilities of 17.4% and 33.9% are close to the sample volatilities of 17.9% and 32.1%. On the standardized residuals, $Q(20)$ of the squares falls to 25.2 ($p = 0.19$) for the market and 11.6 ($p = 0.93$) for MSFT, so one GARCH(1,1) equation captures essentially all of the clustering. It does not capture the tails, since the residual kurtosis is still 4.9 and 8.8. The Gaussian likelihood is therefore a quasi-likelihood: the estimates remain consistent if the mean and variance equations are right, but the classical standard errors are not valid. For MSFT the robust standard errors are two to three times larger.

(c) Is MSFT's volatility just the market moving?

If the risk manager were right, MSFT's conditional variance would be close to $b^2\sigma_{m,t}^2$, and three things would follow. The market would explain most of MSFT's return variance, the firm-specific residual would carry no volatility dynamics of its own, and MSFT's variance would rise and fall one-for-one with the market's. The estimates reject all three.

The market model $r_{MSFT,t} = a + b\,r_{m,t} + u_t$ gives $\hat{b} = 1.15$ (White s.e. 0.021) and $R^2 = 0.41$. Over 1990–2020 the market accounts for 41% of MSFT's daily variance, and the other 59% is firm-specific. The residual is also far from noise. A GARCH(1,1) fitted to u_t gives $\alpha = 0.024$ (0.006) and

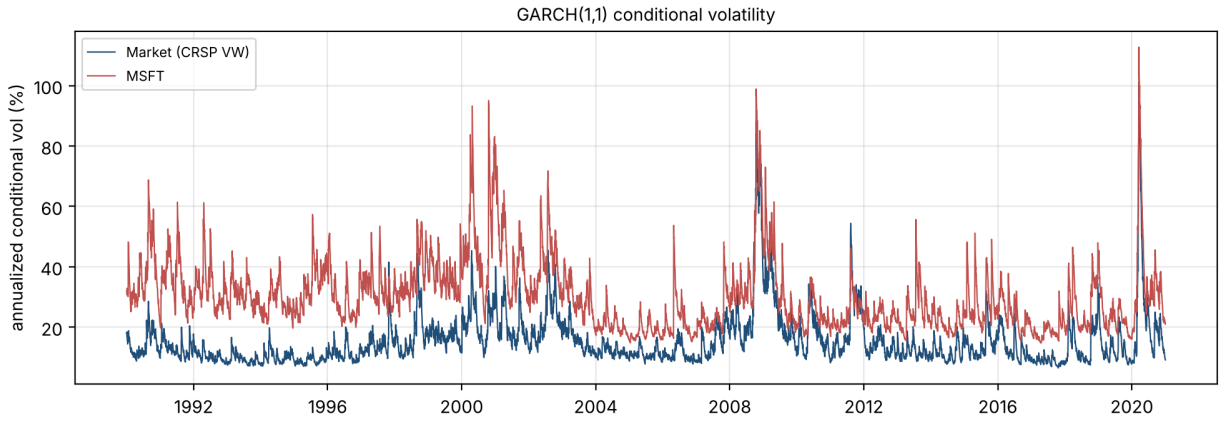


Figure 2: GARCH(1,1) conditional volatility, annualized.

$\beta = 0.971$ (0.007), with persistence 0.995 (0.002), and $Q(20)$ of u_t^2 is 704. Firm-specific volatility therefore has its own slow-moving state, at least as persistent as the market's, and a beta-scaled market forecast would miss it entirely. Finally, a regression of MSFT's fitted log variance on the market's fitted log variance has a slope of 0.50 and an R^2 of 0.39. Under the claim both would be close to one. A doubling of market variance is associated with roughly a 40% rise in MSFT's variance, and most of the movement in MSFT's variance is unrelated to the market's.

The two GARCH fits also show when the claim comes closest to holding. Combining them gives an approximate market-related share of MSFT's variance,

$$s_t = \frac{b^2 \hat{\sigma}^2_{m,t}}{b^2 \hat{\sigma}^2_{m,t} + \hat{\sigma}^2_{u,t}}.$$

This is an approximation, not an exact conditional decomposition, because it assumes a constant beta and a market return that is conditionally uncorrelated with u_t. OLS only guarantees zero correlation on average, and yearly betas range from 0.73 to 2.30. As a check, the implied total variance averages 1.05 times MSFT's own GARCH variance, with a 10th–90th percentile range of 0.72 to 1.43. The share averages 0.36 and exceeds one half on only 20% of days.

Period	Approx. market share	Market-related vol	Firm-specific vol
Full sample 1990–2020	0.36	17.8%	23.5%
Dot-com and antitrust 2000–02	0.36	25.3%	33.8%
Financial crisis Sep 2008–Jun 2009	0.64	49.4%	33.7%
COVID crash Mar–Apr 2020	0.87	71.3%	24.2%
Calm year 2017	0.32	10.0%	14.7%

Table 2: Average approximate market-related share of MSFT's conditional variance, and the average annualized volatility of each part.

The share is high only in market-wide crises. In the 2008–09 financial crisis it averaged 0.64 and in March–April 2020 it reached 0.87, because market volatility exploded while MSFT's firm-specific volatility rose much less. In those months the risk manager's description is roughly right. In calm markets it is wrong (0.32 in 2017), and in 2000–02 it is wrong for the opposite reason: MSFT's volatility was very high, but the antitrust case and the end of the technology boom raised its firm-specific volatility (33.8%) as much as its market-related volatility, so the share stayed at its average.

The relationship has tightened in recent years, with a yearly R^2 of 0.64 to 0.75 in 2018–2020, as Microsoft became one of the largest weights in the index. Taken over the whole sample, though, most of MSFT's risk is its own, and a risk model that treats MSFT as leveraged market exposure would understate its risk in normal times and miss firm-specific episodes altogether.

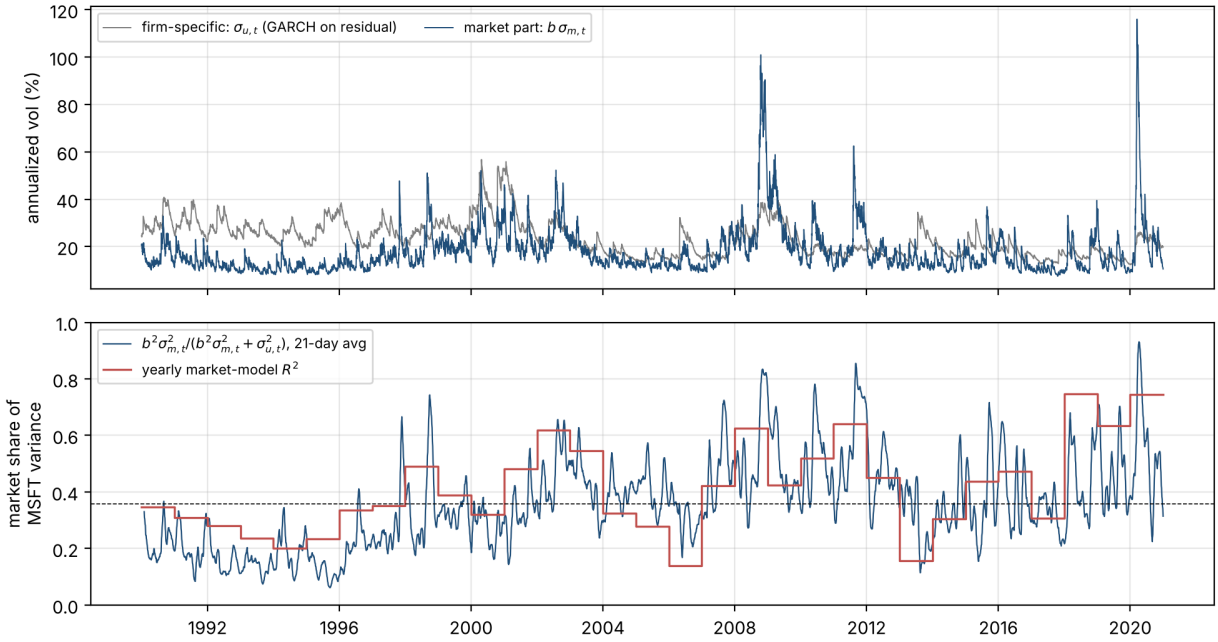


Figure 3: Top: market-related ($b\,\sigma_{m,t}$) and firm-specific ($\sigma_{u,t}$) volatility of MSFT. Bottom: approximate market-related share of MSFT's variance (dashed: mean) and yearly market-model R^2.

(d) Market risk over the next month (as of 31 December 2020)

The forecasts come from the fitted market model. The variance forecast h days ahead is $\mathbb{E}_t\sigma^2_{t+h} = \bar{\sigma}^2 + (\alpha+\beta)^{h-1}(\sigma^2_{t+1} - \bar{\sigma}^2)$, and the one-month variance is the sum of the 21 daily forecasts, which gives a one-month standard deviation of 3.0% against 5.0% at the long-run level. A month of GARCH returns is not exactly normal even with normal daily shocks, so the ranges and loss probabilities below come from 200,000 simulated 21-day paths of the fitted model. The explanation for a reader who has never heard of GARCH comes down to three points.

First, market risk is low right now. The model reads the size of recent daily moves and concludes that the market is currently running at about 9% volatility a year, roughly half its long-run average of 17%. The last weeks of 2020 were quiet, and quiet periods in the stock market tend to last.

Second, risk should creep back toward normal, but slowly. The distance between today's risk and the long-run level, measured in variance, shrinks by half about every two months, so by the end of January the model's estimate of annual volatility has only risen to about 12%. In practice this means a typical month ahead moves the market by about 3% up or down, against about 5% in an ordinary month. In 19 of 20 simulated months the market ends somewhere between a 4.5% loss and a 7.5% gain, and a loss of more than 6% happens in about 1 month out of 100.

Third, these numbers can change in a single day. The forecast is driven by the latest moves, so one large drop would raise it immediately. On 17 March 2020 the same model put annual volatility

near 100%. The model also treats daily surprises as bell-shaped, while real market returns have more extreme days than that, so the true chance of a crash is higher than these figures suggest.

Question 2 (PhD): Gaussian GARCH-in-mean

In the GARCH-in-mean model the conditional variance enters the expected return, $r_t = \mu + \lambda\, g(\sigma_t^2) + \varepsilon_t$ with $\varepsilon_t = \sigma_t z_t$, $z_t \sim N(0,1)$ and $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, where $g(\sigma^2)$ is usually σ^2, σ or $\log\sigma^2$. The MLE solves

$$\hat\theta = \arg\max_\theta \; -\frac{1}{2}\sum_{t=1}^{T}\left[\log 2\pi + \log\sigma_t^2(\theta) + \frac{\big(r_t - \mu - \lambda g(\sigma_t^2(\theta))\big)^2}{\sigma_t^2(\theta)}\right] \quad \text{s.t. } \omega > 0,\ \alpha,\beta \geq 0,\ \alpha+\beta < 1.$$

The difference from plain GARCH is that the residual $\varepsilon_t = r_t - \mu - \lambda g(\sigma_t^2)$ now depends on the variance parameters, so the mean and variance recursions have to be run together: σ_t^2 is built from ε_{t-1}, and ε_t is then built from σ_t^2. For the same reason the information matrix is no longer block-diagonal between mean and variance parameters. A misspecified variance equation now biases the estimate of λ, and the two blocks cannot be estimated in separate steps. With $g = \sigma^2$ we estimate $\hat\lambda = 0.041$ (0.013) for the market and 0.012 (0.009) for MSFT, a positive risk–return trade-off for the market and none for the single stock, which fits the idea that only systematic risk is priced.

Additional return predictors enter the mean equation. With $m_t = \mu + \lambda g(\sigma_t^2) + \gamma' X_{t-1}$ the objective is unchanged except that $\varepsilon_t = r_t - m_t$, and the variance recursion is fed this new residual. The predictors must be known at $t-1$, and they must not be collinear with $g(\sigma_t^2)$. If one of them is itself a volatility proxy, such as a lagged VIX, λ and γ are only weakly identified.

Variance predictors enter the GARCH equation, $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 + \delta' Q_{t-1}$, again with the same objective, but the restrictions that make plain GARCH well behaved no longer cover the new term. The first problem is positivity. The conditions $\omega > 0$ and $\alpha, \beta \geq 0$ guarantee $\sigma_t^2 > 0$ in plain GARCH, but $\delta' Q_{t-1}$ can be negative, and a single negative variance makes the likelihood undefined. The simplest fix is to use non-negative predictors, such as variances or squared spreads, together with $\delta \geq 0$. For predictors of either sign one has to restrict $\omega + \delta' Q_t > 0$ over the support of Q or reparametrize, for example with $\exp(\delta' Q_{t-1})$ or an EGARCH-X model for $\log\sigma_t^2$. The predictors must also be predetermined, $Q_{t-1} \in \mathcal{F}_{t-1}$; otherwise σ_t^2 is no longer a conditional variance and the prediction-error decomposition behind the likelihood breaks down.

The second problem is stationarity. With an exogenous input, $\alpha + \beta < 1$ is no longer enough, because the variance inherits the behaviour of Q. Q_t must be strictly stationary and ergodic with $\mathbb{E}|Q_t| < \infty$, in which case $\mathbb{E}\sigma^2 = (\omega + \delta'\mathbb{E}Q)/(1-\alpha-\beta)$; a trending or unit-root predictor would make σ_t^2 non-stationary however small $\alpha + \beta$ is. These conditions, together with identification, give consistency of the (Q)MLE. Asymptotic normality needs more: an interior true parameter, $\mathbb{E}z_t^4 < \infty$ so that the score has finite variance, and enough moments of Q_t and σ_t^2 for the score and Hessian to obey a law of large numbers and a central limit theorem. Identification requires that Q vary over time and not be collinear with ε_{t-1}^2 or σ_{t-1}^2, since otherwise ω, δ and β cannot be separated. Finally, the sign restriction matters for testing. If $\delta \geq 0$ is imposed, $\delta = 0$ lies on the boundary of the parameter space, and for a scalar δ with the other parameters interior the likelihood-ratio statistic converges to $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ rather than χ_1^2.

Question 3 (PhD): GJR-GARCH(1,1,1) with Student-t innovations

The model is $r_t = \mu + \varepsilon_t$, $\varepsilon_t = \sigma_t z_t$, with z_t i.i.d. standardized Student-t (mean 0, variance 1, $\nu > 2$) and

$$\sigma_t^2 = \omega + (\alpha + \gamma\,\mathbf{1}\{\varepsilon_{t-1} < 0\})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2,$$

so a negative return raises next-day variance by $\alpha + \gamma$ per unit of squared return and a positive return by α. The log likelihood is

$$\mathcal{L} = \sum_t \left[\log\Gamma\!\left(\tfrac{\nu+1}{2}\right) - \log\Gamma\!\left(\tfrac{\nu}{2}\right) - \tfrac{1}{2}\log\left(\pi(\nu-2)\right) - \tfrac{1}{2}\log\sigma_t^2 - \tfrac{\nu+1}{2}\log\!\left(1 + \frac{\varepsilon_t^2}{(\nu-2)\sigma_t^2}\right)\right].$$

To derive the non-integration condition, substitute $\varepsilon_{t-1}^2 = \sigma_{t-1}^2 z_{t-1}^2$ and note that $\mathbf{1}\{\varepsilon_{t-1} < 0\} = \mathbf{1}\{z_{t-1} < 0\}$ because $\sigma_{t-1} > 0$:

$$\sigma_t^2 = \omega + \sigma_{t-1}^2\big[\alpha z_{t-1}^2 + \gamma z_{t-1}^2 \mathbf{1}\{z_{t-1} < 0\} + \beta\big].$$

z_{t-1} is independent of $\sigma_{t-1}^2 \in \mathcal{F}_{t-2}$, $\mathbb{E}z^2 = 1$, and because the standardized t is symmetric about zero, $\mathbb{E}[z^2\mathbf{1}\{z<0\}] = \frac{1}{2}\mathbb{E}z^2 = \frac{1}{2}$. Taking expectations at $t-2$,

$$\mathbb{E}_{t-2}\sigma_t^2 = \omega + \phi\,\sigma_{t-1}^2, \qquad \phi = \alpha + \tfrac{1}{2}\gamma + \beta,$$

and iterating gives $\mathbb{E}_{t-k}\sigma_t^2 = \omega\sum_{j=0}^{k-2}\phi^j + \phi^{k-1}\sigma_{t-k+1}^2$. The effect of today's variance on the forecast k periods ahead decays like ϕ^k, and the unconditional variance $\mathbb{E}\varepsilon_t^2 = \omega/(1-\phi)$ is finite if and only if

$$\boxed{\alpha + \tfrac{1}{2}\gamma + \beta < 1} \qquad (\omega > 0,\ \alpha \geq 0,\ \alpha + \gamma \geq 0,\ \beta \geq 0,\ \nu > 2).$$

At $\phi = 1$ the variance is integrated, shocks to it never die out, and the unconditional variance does not exist. The leverage term enters with weight one half because, for a symmetric shock distribution, negative shocks account for half of $\mathbb{E}z^2$.

	μ	ω	α	γ	β	ν	ϕ	$\log L$
Market	0.065	0.0137	0.009	0.163	0.895	6.9	0.985	−9,832.4
	(0.008)	(0.0024)	(0.006)	(0.018)	(0.010)	(0.57)	(0.003)	
MSFT†	0.073	0.032	0.056	0.068	0.910	5.3	0.99999	−15,220.2
	(0.016)	(0.010)	(0.010)	(0.015)	(0.014)	(0.32)	–	

Table 3: GJR-GARCH-t estimates; robust s.e. in parentheses. †The constraint $\phi \leq 1 - 10^{-5}$ imposed in estimation binds for MSFT ($\hat{\phi} = 0.99999$). Without it $\hat{\phi} = 1.0017$ with a log-likelihood gain of 0.12 (LR = 0.24). The MSFT standard errors are computed as if the estimate were interior and are only indicative.

	GARCH-N	GARCH-t	GJR-N	GJR-t
Market	−10,110.2	−9,927.5	−9,999.7	−9,832.4
MSFT	−15,649.6	−15,238.1	−15,624.0	−15,220.2

Table 4: Log likelihood by model, showing how much fat tails and leverage each add.

Relative to the Gaussian GARCH(1,1), GJR-t raises the log likelihood by 278 for the market and 429 for MSFT, far more than two extra parameters could produce by chance. The second table

shows where the gain comes from. Fat tails matter for both series ($\hat{\nu} = 6.9$ and 5.3; switching to t errors alone adds 183 and 412), and they matter most for MSFT, whose firm-specific jumps are exactly the extreme standardized shocks a normal distribution cannot produce. Leverage matters a great deal for the market (it adds 111) and little for MSFT (25).

The parameters move in ways that follow from this. For the market, the symmetric coefficient falls from 0.112 to an insignificant 0.009 and the asymmetry term $\gamma = 0.163$ takes over, so the GARCH(1,1) α was an average of a strong response to bad news and almost no response to good news. Market persistence is unchanged ($\phi = 0.985$, s.e. 0.0035, $t = 4.3$ against one), and the long-run volatility falls from 17.4% to 15.2% because the asymmetric model explains high-variance periods as runs of negative shocks instead of a higher baseline level. For MSFT persistence rises to the bound $\phi \leq 1 - 10^{-5}$. Under t errors the largest firm-specific moves are treated as draws from a fat tail, so they no longer pull β down, and the fitted variance becomes smoother and more persistent. Dropping the bound gives $\hat{\phi} = 1.0017$ with a log-likelihood gain of only 0.12, so the data cannot distinguish MSFT's variance in this model from an integrated process, its long-run variance is not identified, and its standard errors rest on interior asymptotics that do not apply at the boundary.

The leverage effect is far stronger for the market. Bad news raises the market's next-day variance 19 times as much as good news of the same size ($\alpha + \gamma = 0.171$ against $\alpha = 0.009$), while for MSFT the ratio is 2.2 (0.124 against 0.056). This is hard to reconcile with the financial-leverage explanation, in which a falling stock price raises the debt-to-equity ratio and therefore equity volatility. Microsoft had no long-term debt until 2009 and little relative to its market value afterwards, yet the asymmetry is weak for MSFT and very strong for the index. A systematic channel fits better. Under volatility feedback, news that raises expected market volatility raises required returns and pushes prices down at the same moment, and correlations across stocks rise in selloffs, so falling prices and rising volatility arrive together in the aggregate. Firm-specific news works differently, because a large earnings surprise raises uncertainty about the firm whether it is good or bad. In the fitted model, the +19.6% move on 19 October 2000 lifted MSFT's annualized volatility from 47% to 86% the next day. MSFT's γ is 0.42 times the market's, close to the 0.36–0.41 share of its variance that is market-related, which is roughly what one would expect if its asymmetry came almost entirely through its market exposure.

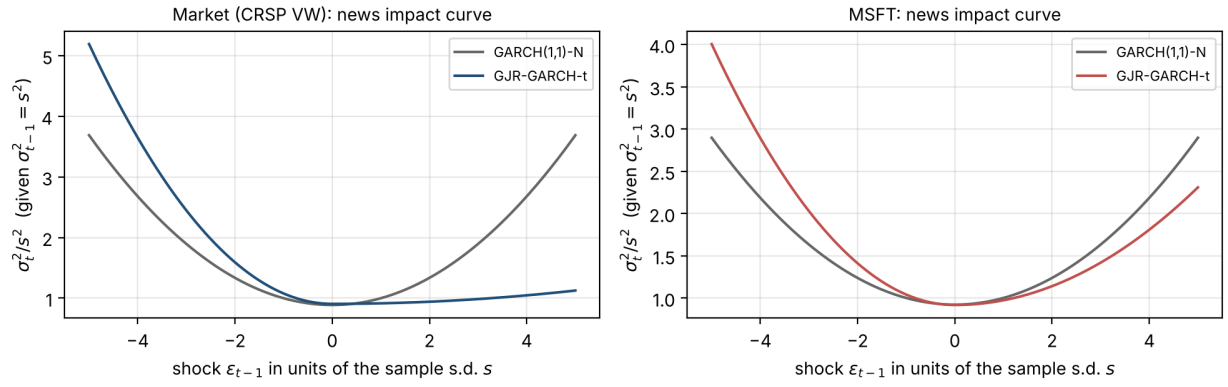


Figure 4: News impact curves: next-day variance after a shock ε_{t-1}, given that the previous variance equals the sample variance s^2. Axes are scaled by the sample s.d. because the long-run variance does not exist for MSFT's GJR-t estimate.